

Def. The Extension field K/F is called a splitting field for $f(x) \in F[x]$ if:

- $\begin{cases} f(x) \text{ splits completely over } K \\ f(x) \text{ does not split completely over any proper subfield of } K \text{ containing } F. \end{cases}$

Thm. Let F be a field, and $f(x) \in F[x]$ be a non-constant polynomial.

There exists a splitting field for $f(x)$.

Proof. Using mathematical Induction for degree of $f(x)$.

If $n=1$, then put $K=F$ be a splitting field.

Now, suppose that $n > 1$, and the statement holds for $\deg \leq n-1$.

Let $\deg f = n$. Denote $f(x)$ as irreducible decomposition form:

$$f(x) = p_1(x) \cdots p_r(x) \text{ where } p_i(x) \in F[x], \quad r \leq n.$$

If for all $1 \leq i \leq r$, $\deg p_i = 1$, then i , all p_i 's are linear factors,

then take $K=F$. Otherwise, at least one $1 \leq i \leq r$, $\deg p_i \geq 2$.

Let E_1 be an Extension of F which has a root α of $p_1(x)$.

Then, $f(x)$ has a linear factor $(x-\alpha)$ in E_1 . Now,

$$f_1(x) = f(x) \cdot (x-\alpha)^{-1} \in E_1[x]$$

has degree less than or equals to $n-1$,

thus there exists an Extension E/E_1 which has all roots of $f_1(x)$.

Since $\alpha \in E_1 \subseteq E$, E has all roots of $f(x)$. Now, since intersection of fields is field

$$K = \bigcap_{F \subseteq H \subseteq E} H$$

H has all roots of $f(x)$

is the splitting field for $f(x)$.

If K/F is an Algebraic Extension field and

which is the splitting field for a collection of polynomials in $F[x]$

then K is called Normal Extension of F .

⌈ Cyclotomic Field: Splitting field of $x^n - 1$.

Consider a field $\mathbb{Q}$ and a polynomial $x^n - 1 \in \mathbb{Q}[x]$.

Since $x^n = 1 \Leftrightarrow x = \exp\left(2\pi i \cdot \frac{k}{n}\right) \quad k = 0, 1, 2, \dots, n-1$

Clearly, $\{x \in \mathbb{C} \mid x^n = 1\}$ is group under the multiplication,

which is isomorphic to $\mathbb{Z}_n$ by an Isomorphism [Note: $\exp(i \cdot \theta) = \cos \theta + i \cdot \sin \theta$

$$\mathbb{Q}[\mathbb{Z}_n] \rightarrow \{x \in \mathbb{C} \mid x^n = 1\} : k \mapsto \exp\left(2\pi i \cdot \frac{k}{n}\right)$$

Since for $a \in \mathbb{Z}_n$,

$$\langle a \rangle = \mathbb{Z}_n \Leftrightarrow \gcd(a, n) = 1$$

Thus, for $k = 0, 1, \dots, n-1$,

$$\langle \exp\left(2\pi i \cdot \frac{k}{n}\right) \rangle = \{x \in \mathbb{C} \mid x^n = 1\} \Leftrightarrow \gcd(k, n) = 1$$

This group has order of $\phi(n)$, where ϕ is Euler function.

Thm. Let $k \in \mathbb{Z}_n$ be given satisfying $\gcd(k, n) = 1$.

Then, $\mathbb{Q}\left(\exp\left(2\pi i \cdot \frac{k}{n}\right)\right)$ is splitting field for $x^n - 1 \in \mathbb{Q}[x]$.

(It is called the cyclotomic field of n^{th} roots of unity.)

Proof. Since $\exp\left(2\pi i \cdot \frac{k}{n}\right)$ can generate all root of $x^n - 1$,

$x^n - 1$ splits over $\mathbb{Q}\left(\exp\left(2\pi i \cdot \frac{k}{n}\right)\right)$. And suppose that K is a field s.t.

$$\mathbb{Q}\left(\exp\left(2\pi i \cdot \frac{k}{n}\right)\right) \supseteq K \supseteq \mathbb{Q}$$

and $x^n - 1$ splits over K . Then, K must be larger than $\mathbb{Q}$,

and must contains at least one generator of $\{x \in \mathbb{C} \mid x^n = 1\}$.

Denote $\zeta_n = \exp\left(2\pi i \cdot \frac{1}{n}\right)$.

Thm $[\mathbb{Q}\left(\exp\left(2\pi i \cdot \frac{1}{n}\right)\right) : \mathbb{Q}] = \phi(n)$

Proof. If n is a prime p , then

$$\frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \dots + 1$$

is irreducible and has ζ_p as a root.

Splitting field of $x^p - 2$, where p is a prime.

Consider a set $\{x \in \mathbb{C} \mid x^p - 2 = 0\}$. Clearly, it is Not a group.

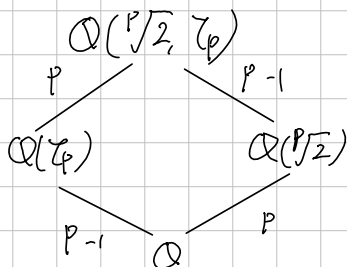
Since for each $0 \leq k \leq p-1$,

$$(\zeta_p^k \cdot \sqrt[p]{2})^p = (\zeta_p^k)^p \cdot 2 = 2 \quad \text{where } \zeta_p = \exp(2\pi i \cdot \frac{1}{p})$$

Therefore,

$$\{x \in \mathbb{C} \mid x^p - 2 = 0\} = \{\zeta_p^k \cdot \sqrt[p]{2} \mid 0 \leq k \leq p-1\}$$

This implies $\mathbb{Q}(\sqrt[p]{2}, \zeta_p)$ is a splitting field of $x^p - 2$.



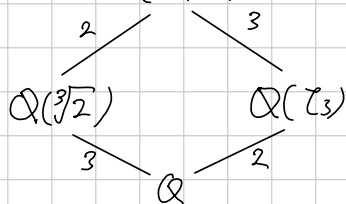
Splitting field of x^3-2 over $\mathbb{Q}$.

Note: $\zeta_3 = \exp(2\pi i \cdot \frac{1}{3}) = \cos \frac{2}{3}\pi + i \sin \frac{2}{3}\pi = -\frac{1}{2} + i \cdot \frac{\sqrt{3}}{2} = \frac{-1 + i\sqrt{3}}{2}$

Thus, the roots of x^3-2 are

$$\sqrt[3]{2}, \quad \sqrt[3]{2} \cdot \frac{-1 + i\sqrt{3}}{2}, \quad \sqrt[3]{2} \cdot \frac{-1 - i\sqrt{3}}{2}$$

Clearly, $\mathbb{Q}(\sqrt[3]{2}, \zeta_3)$ is a splitting field of x^3-2 ,



Consider $\mathcal{I}: \mathbb{Q} \rightarrow \mathbb{Q}: x \mapsto x$, Automorphism.

Let $\bar{\mathcal{I}}$ be an Extension of $\mathcal{I}$ to $\mathbb{Q}(\sqrt[3]{2})$ into $\bar{\mathbb{Q}}$.

Then, $\bar{\mathcal{I}}(\sqrt[3]{2})$ must be a root of x^3-2 . Thus is, the possible cases are:

$$\bar{\mathcal{I}}: \mathbb{Q}(\sqrt[3]{2}) \rightarrow \bar{\mathbb{Q}}: a_0 + a_1 \sqrt[3]{2} \mapsto a_0 + a_1 \cdot \sqrt[3]{2} \cdot \zeta_3^k \quad (k=0,1,2).$$

Thm. Let $\varphi: F \xrightarrow{\sim} F'$ be a field Isomorphism.

Let $f(x) = \sum_{i=0}^n \alpha_i x^i \in F[x]$ be a polynomial, and $f'(x) = \sum_{i=0}^n \varphi(\alpha_i) x^i \in F'[x]$.

Let E, E' be splitting fields for f, f' over F, F' , respectively.

Then, there exists an Isomorphism $\Phi: E \rightarrow E'$ s.t. $\Phi|_F = \varphi$

Proof. Using Induction for degree $n \geq 1$ of the given polynomial.

If $n=1$, then $E=F$, so there is nothing to prove.

Now, suppose that for $\deg f < n$, the statement holds.

Let $p(x) \in F[x]$ be an irreducible factor of $f(x)$, with $\deg p \geq 2$

(If no such p exists, then $E=F$, nothing to prove.)

Let $p'(x) \in F'[x]$ be an irreducible factor of $f'(x)$ corresponding to $p(x)$.

If $\alpha \in E$ is a root of $p(x)$ and $\beta \in E'$ is root of $p'(x)$, then

$$\begin{aligned} \varphi': F[x]/(p(x)) &\rightarrow F'[x]/(p'(x)) : g(x) + (p(x)) \mapsto g'(x) + (p'(x)) \\ &\quad \underbrace{= F_1 \cong F(\alpha)}_{F_1 \cong F(\alpha)} \quad \underbrace{F'_1 \cong F(\beta)}_{F'_1 \cong F(\beta)} \end{aligned}$$

(where $g'(x)$ is obtained from taking φ to all coefficients of $g(x)$)

is an Isomorphism. We have that:

$$f(x) = (x-\alpha) f_1(x), \quad f'(x) = (x-\beta) f'_1(x)$$

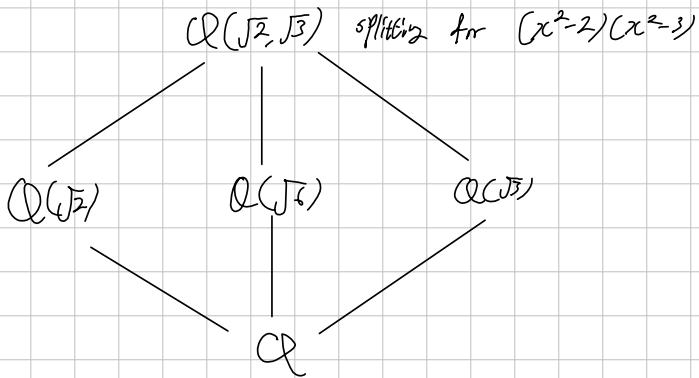
over F_1 and F'_1 , respectively. Now, f_1, f'_1 have degree $\leq n-1$, and

E, E' are splitting fields of f, f' , respectively. (If not, it is contradict to minimality.)

Now, there exists an Extension

$$\Phi: E \rightarrow E' \quad \text{s.t.} \quad \Phi|_{F_1} = \varphi'$$

Q. Let F be a field $f(x) \in F[x]$.



Def. Let $f(x) \in F[x]$ be given as:

$$f(x) = a_n x^n + \dots + a_1 x + a_0 = \sum_{i=0}^n a_i x^i \in F[x].$$

define derivative of $f(x)$ as:

$$D_x f(x) = n \cdot a_n x^{n-1} + \dots + 2 \cdot a_2 x + a_1 = \sum_{i=1}^n i \cdot a_i x^{i-1} \in F[x].$$

Lemma. Let $f, g \in F[x]$. Then,

$$D_x(f(x) + g(x)) = D_x f(x) + D_x g(x)$$

$$D_x(f(x)g(x)) = f(x)D_x g(x) + (D_x f(x))g(x)$$

Thm. Suppose $f(x) \in F[x]$ has a root α in the splitting field E . Then, $f(x) \in F[x]$ has a multiple root α iff $D_x f(\alpha) = 0$.

Proof. Suppose that $f(x)$ has a multiple root α in the splitting field E . Then,

$$f(x) = (x - \alpha)^2 \cdot g(x)$$

where $g(x) \in E[x]$. By the Lemma,

$$D_x f(x) = (x - \alpha)g(x) + (x - \alpha)^2 D_x g(x)$$

Clearly has α as a root.

Conversely, since α is not of $f(x)$, it is represented in E as

$$f(x) = (x - \alpha)h(x)$$

Now, by assumption,

$$0 = D_x f(\alpha) = h(\alpha) + (\alpha - \alpha)D_x f(\alpha) = h(\alpha)$$

Hence $h(x) = (x - \alpha)h_1(x)$ for some h_1 . In sum,

$$f(x) = (x - \alpha)^2 h_1(x).$$

Corollary. Let F be a field s.t. $\text{ch}(F) = 0$. Then,

Every irreducible polynomial in F is separable.

Moreover,

$f(x) \in F[x]$ is separable if and only if $f(x)$ is product of distinct irreducible polynomials.

Proof. Let $p(x) \in F[x]$ be an irreducible. Then, $\gcd(p(x), D_x p(x)) = 1$.

$$x^2 + 4x + 4$$

$$\Rightarrow 2x + 4 = 2(x+2)$$

$$a = \sigma^{-r}(b) \text{ and}$$

$$\sum_{n=0}^{\infty} x_n \text{ converges } (x_n \geq 0)$$

$$\Rightarrow \sum_{n=0}^{\infty} \frac{\sqrt{x_n}}{n} \text{ converges.}$$